

Lecture 09: band structures; DOS; metals and insulators

Prof. Daniel Schwalbe-Koda

MAT SCI 120 — Winter 2026



Samueli
School of Engineering

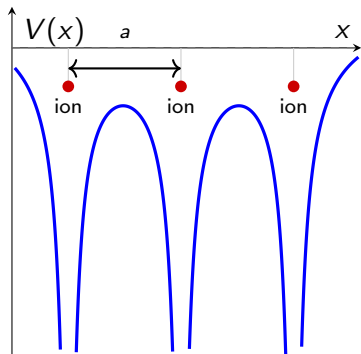
Recap: Introducing a periodic lattice potential

- ▶ Real crystals have periodic arrangement of ion cores
- ▶ Electrons experience attractive Coulomb wells at each lattice site
- ▶ Periodicity condition:

$$V(x + a) = V(x)$$

where a is the lattice constant

- ▶ This periodic potential fundamentally modifies allowed electron energies



Problem

How do we solve the Schrödinger's equation for a periodic potential?

Recap: Bloch's theorem

Bloch wave function

For a periodic potential $V(x) = V(x + a)$, wave functions have the form:

$$\psi_k(x) = u_k(x) e^{ikx}$$

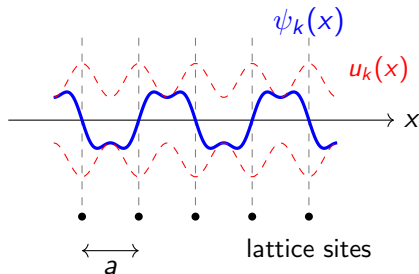
Notation:

- ▶ $u_k(x)$: periodic function with lattice period, $u_k(x) = u_k(x + a)$
- ▶ e^{ikx} : plane wave describing propagation
- ▶ k : wave vector (crystal momentum $= \hbar k$)

Key point: Wave vector k labels different quantum states in the crystal.

Recap: Physical meaning of Bloch waves

- ▶ Plane wave e^{ikx} describes propagation through crystal
- ▶ Periodic part $u_k(x)$ modulates amplitude near each atom
- ▶ Probability density $|\psi|^2$ is periodic (unit cells are equivalent)
- ▶ Bloch state means that electrons are not free nor localized



Recap: consequences of the Bloch's theorem

Translation property

Shifting by one lattice constant multiplies ψ by a phase factor:

$$\psi_k(x + a) = e^{ika} \psi_k(x)$$

Important consequences:

- ▶ Wave function is **not** periodic, but probability density **is**:

$$|\psi_k(x + a)|^2 = |e^{ika}|^2 |\psi_k(x)|^2 = |\psi_k(x)|^2$$

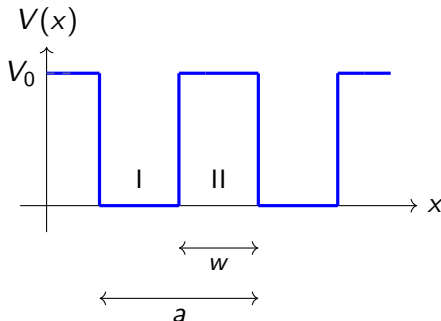
- ▶ All relevant physics is within the period $2\pi/a$ in k -space
- ▶ All relevant k is within the **first Brillouin zone**: $-\frac{\pi}{a} \leq k \leq \frac{\pi}{a}$

Recap: Kronig-Penney model

- ▶ Replace complex, periodic Coulomb potential with periodic square wells
- ▶ Well depth V_0 , barrier width w
- ▶ Lattice constant: a
- ▶ Solve Schrödinger equation in each region, apply Bloch's theorem

Boundary condition

$$\psi(x + a) = e^{ika}\psi(x)$$



Recap: solving the Schrödinger equation for the Kronig-Penney model

Notation: α = wave vector in well, β = decay constant in barrier, m = electron mass

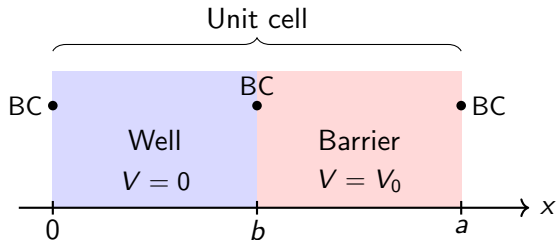
Solutions in each region

- ▶ In well, region I ($V = 0$): $\psi = Ae^{i\alpha x} + Be^{-i\alpha x}$
- ▶ In barrier, region II ($V = V_0$): $\psi = Ce^{\beta x} + De^{-\beta x}$

$$\alpha = \sqrt{\frac{2mE}{\hbar^2}}, \quad \beta = \sqrt{\frac{2m(V_0 - E)}{\hbar^2}} \quad (1)$$

- ▶ Four unknowns (A, B, C, D) require four boundary conditions
- ▶ Continuity of ψ and $d\psi/dx$ at each interface

Recap: Applying boundary conditions (BC) considering periodicity



- ▶ Match ψ and $d\psi/dx$ at $x = b$ (well-barrier interface)
- ▶ Match ψ and $d\psi/dx$ at $x = a$ (barrier-well interface)
- ▶ Bloch condition: $\psi(a) = e^{ika}\psi(0)$ relates solutions across unit cell
- ▶ Nontrivial solution (i.e., $\psi = 0$ is not useful)

The Kronig-Penney dispersion relation

Notation: w = barrier width, a = lattice parameter

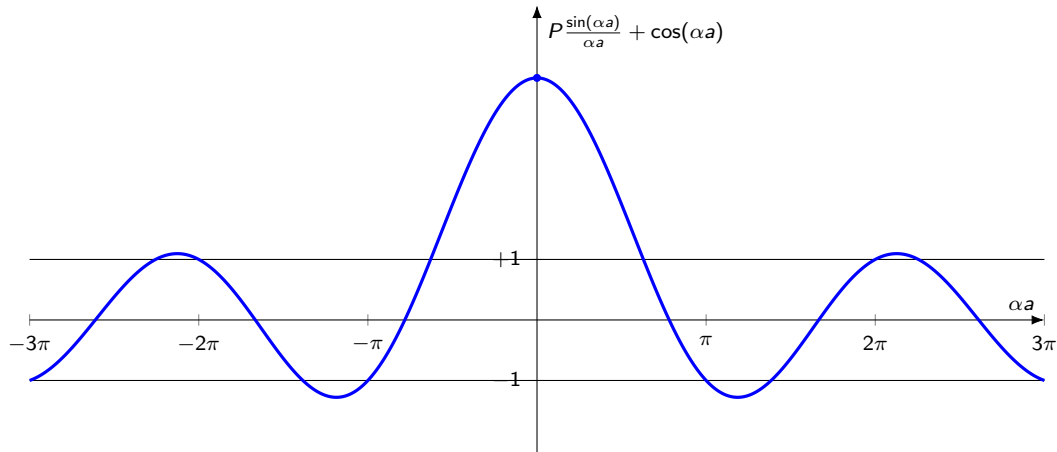
Kronig-Penney Equation

$$\cos(ka) = P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) \quad (2)$$

where $P = \frac{ma}{\hbar^2} V_0 w$ and $\alpha = \sqrt{\frac{2mE}{\hbar^2}}$

- ▶ Left-hand side ($\cos ka$) is bounded: $-1 \leq \cos(ka) \leq +1$
- ▶ Right-hand side depends on the geometry of the system, V_0 , and electron energy E
- ▶ Solutions exist only when $P \frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) \leq 1$: **allowed energy bands**

Band gaps emerge at $k = n\pi/a$



Constraints for the Kronig-Penney model

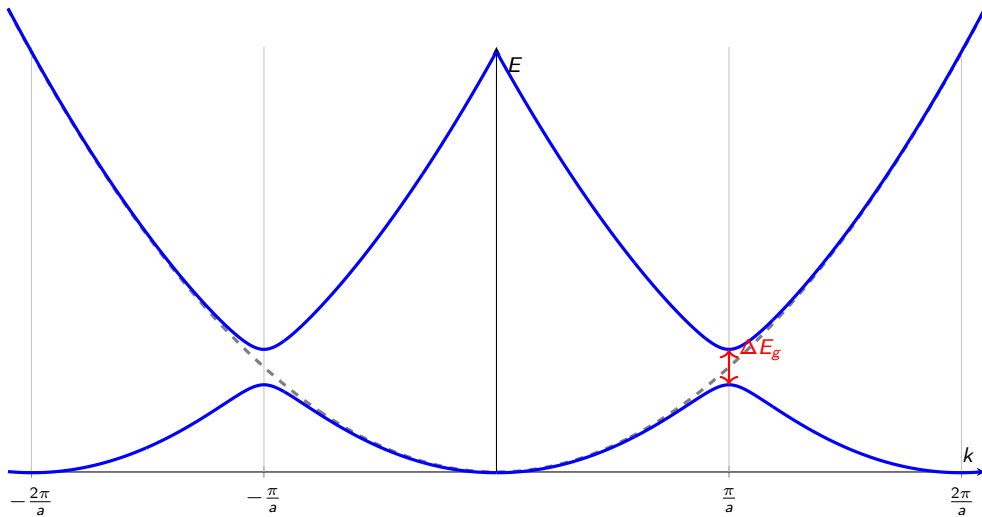
- ▶ $|\cos(ka)| > 1$: no real $k \rightarrow$ **forbidden regions**

- ▶ Gaps happen at Brillouin zone boundaries:

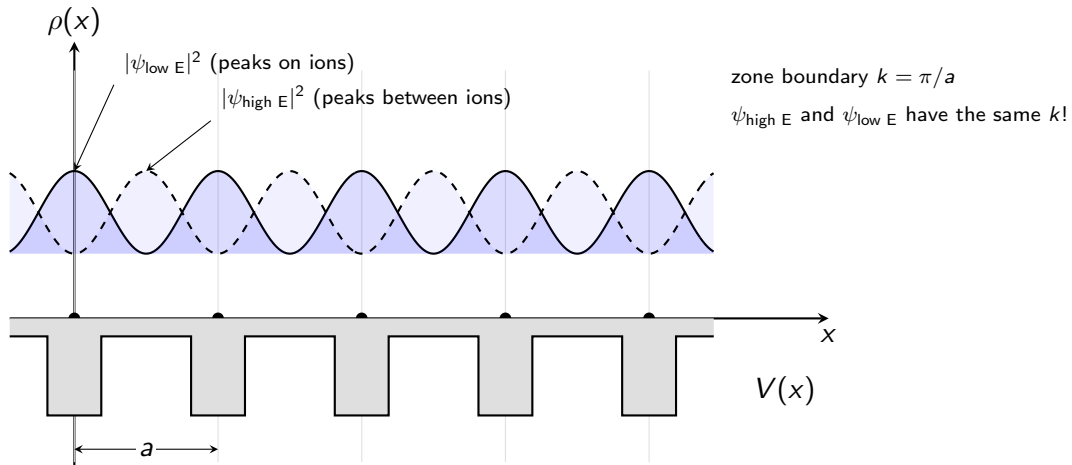
$$k = \frac{n\pi}{a}, \quad n = \pm 1, \pm 2, \dots$$

- ▶ Gaps happen because of the Bragg condition: $2a = n\lambda$
- ▶ Electrons with the same periodicity of the lattice form standing waves with split energies

Gap opening in the Kronig-Penney model



Intuition on why a gap exists (real space wavefunctions $\psi(x)$)



Conceptual questions: Kronig-Penney model summary

Conceptual questions:

- ▶ **Q1:** Why use Bloch's theorem instead of simple periodic boundary conditions?
- ▶ **Q2:** What determines whether energy E is allowed or forbidden?
- ▶ **Q3:** Why do gaps appear specifically at $k = n\pi/a$?
- ▶ **Q4:** Why did we draw multiple lines per k value two slides ago?
- ▶ **Q5:** Why are we dealing with parabolas for energy?
- ▶ **Q6:** Can you use the Kronig-Penney model to explain the trends in electrical behavior in materials? (metallic bonds, covalent bonds, ionic bonds)

Key insight: Periodic potential + quantum mechanics $\rightarrow$ discrete allowed bands

Wave packets in lattices explain particle-like dynamics of electrons

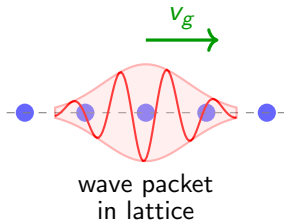
- ▶ Electrons in crystals are **wave packets** with group velocity

$$v_g = \frac{1}{\hbar} \frac{\partial E}{\partial k}$$

- ▶ An external electric field creates a force F on electrons, changing their momentum:

$$\hbar \frac{dk}{dt} = F$$

- ▶ Goal: Connect quantum band structure to classical $F = ma$ dynamics



Deriving effective mass: step 1 – acceleration

Goal: Find acceleration of electron wave packet

Step 1: Acceleration is time derivative of group velocity

$$a = \frac{dv_g}{dt}$$

Step 2: Apply chain rule using $v_g = \frac{1}{\hbar} \frac{\partial E}{\partial k}$

$$a = \frac{d}{dt} \left(\frac{1}{\hbar} \frac{\partial E}{\partial k} \right) = \frac{1}{\hbar} \frac{\partial^2 E}{\partial k^2} \frac{dk}{dt}$$

Step 3: Substitute $\hbar \frac{dk}{dt} = F$ (applied force)

$$\boxed{a = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2} F} \quad (3)$$

Deriving effective mass: step 2 – definition

Compare with Newton's second law:

$$\text{Classical: } a = \frac{F}{m^*} \quad (4)$$

$$\text{Derived: } a = \frac{1}{\hbar^2} \frac{\partial^2 E}{\partial k^2} F \quad (5)$$

Matching coefficients:

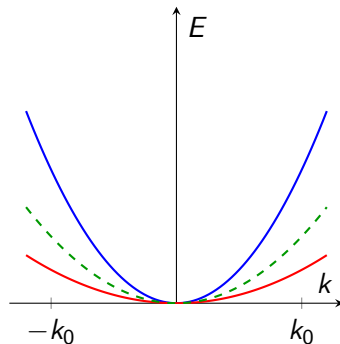
$$m^* = \hbar^2 \left(\frac{\partial^2 E}{\partial k^2} \right)^{-1} \quad (6)$$

- ▶ Effective mass emerges naturally from **band curvature**
- ▶ m^* replaces free electron mass m in Newton's second law
- ▶ Quantum mechanics $\rightarrow$ classical dynamics with modified mass

Band curvature governs electron dynamics

— Small m^* — Large m^* - - - Free e^-

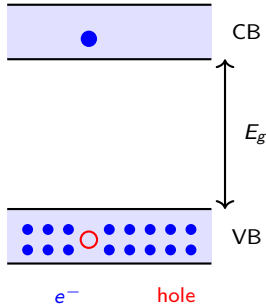
- ▶ **Large curvature** $\rightarrow$ small m^* $\rightarrow$ electrons accelerate easily
- ▶ **Small curvature** (flat band) $\rightarrow$ large m^* $\rightarrow$ sluggish response
- ▶ Free electron: $E = \frac{\hbar^2 k^2}{2m}$ gives $m^* = m$ exactly



Steeper curvature = lighter effective mass = faster response

Holes are missing electrons in bands

- ▶ A **hole** is a missing electron in a nearly-full valence band
- ▶ Hole behaves as particle with charge $+e$
- ▶ Hole effective mass is **positive** (opposite sign to missing electron's m^*)
- ▶ Simplifies analysis: track few holes instead of many electrons



Why holes are equivalent to missing electrons from a current perspective?

Current from a nearly-full band:

Full band has zero current: $\sum_{\text{all } k} \mathbf{v}_k = 0$ (electrons don't have available states to "move")

With one electron missing:

$$\mathbf{J} = -e \sum_{\text{filled}} \mathbf{v}_k = -e \left(\sum_{\text{all}} \mathbf{v}_k - \mathbf{v}_{\text{missing}} \right) = +e \mathbf{v}_{\text{missing}}$$

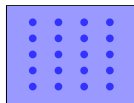
$$\boxed{\mathbf{J}_{\text{hole}} = +e \mathbf{v}_{\text{missing}}}$$

(7)

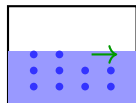
Hole properties:

- ▶ Charge: $q_{\text{hole}} = +e$ (positive!)
- ▶ Mass: $m_{\text{hole}}^* = -m_{\text{electron}}^*$ (positive near VB top)
- ▶ Hole responds to fields as a **positive particle**

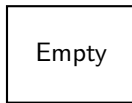
Band filling determines conductivity



Full band
 $\sum v_k = 0$
No current



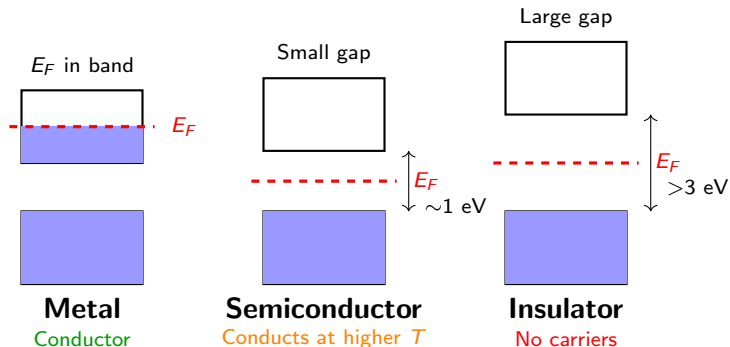
Partial
 $\sum v_k \neq 0$
Conducts!



Empty band
No carriers
No current

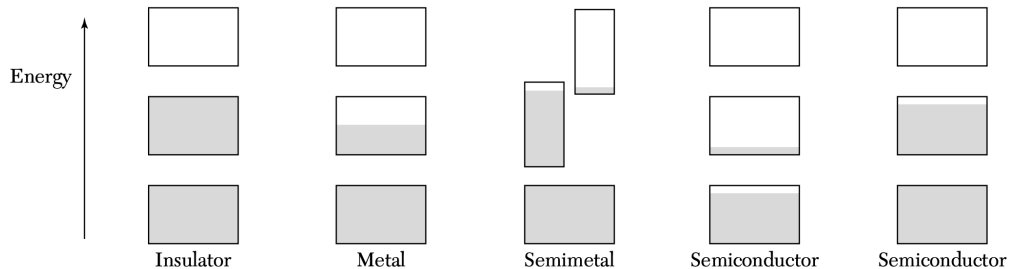
- ▶ **Full bands:** All k -states paired $\Rightarrow \sum v_k = 0$
- ▶ **Partial bands:** Electrons can redistribute $\Rightarrow$ conduction
- ▶ Fermi level E_F indicates which bands participate

Materials and their band diagram



Classification rule: Band filling + gap size $\Rightarrow$ electrical behavior

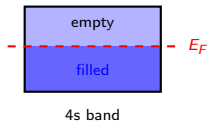
More general picture of band diagrams



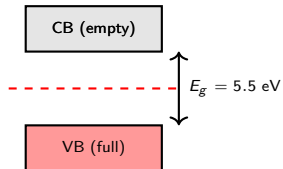
Metals and insulators: Cu vs diamond

- ▶ **Copper:** Partially filled 4s band, E_F inside band $\rightarrow$ excellent conductor
- ▶ **Diamond:** $E_g = 5.5$ eV, $k_B T \ll E_g \rightarrow$ negligible excitation, insulator
- ▶ Band theory explains $\sim 10^{20}$ difference in conductivity!
- ▶ Effective mass in Cu $\approx m_e$; irrelevant in insulators at $T = 300$ K

Copper (Metal)



Diamond (Insulator)



Conceptual questions: holes and material classification

Test your understanding:

1. **Q1:** Why does a full band contribute zero current even with applied field?
2. **Q2:** Why does a hole have a negative effective mass?
3. **Q3:** Can a semiconductor conduct at 0 K? Justify this using the Fermi-Dirac distribution
4. **Q4:** Where is the Fermi level in a metal?
5. **Q5:** Why is the Fermi level in a semiconductor in the middle of the gap?
6. **Q6:** What if $|m_h^*| \ll |m_e^*|$? What do you expect for the carrier that dominates the conduction of this system?